

Reproducing DESI Strong Lens Foundry III: Keck NIRES Source Redshifts for ResNet-Discovered Strong Lenses

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Internal Reproduction Report

Abstract

We reproduce the redshift-measurement core of [Agarwal et al. \(2025\)](#) (DESI Strong Lens Foundry III), which secured near-infrared source redshifts for the eight ResNet-discovered, HST-confirmed strong lenses observed with the Keck Near-Infrared Echellette Spectrometer (NIRES). The paper detects emission lines in six of the eight NIRES sources and reports source redshifts spanning $z_s = 1.675\text{--}3.332$, measured by fitting the published two-Gaussian, six-parameter model (their Eq. 1) to the PYPEIT-reduced 1D spectra. We implement that exact fit model from scratch (`scipy.optimize.curve_fit`) and recover all six published source redshifts blind to $|\Delta z| < 0.001$ (worst single-shot 2.2×10^{-4} ; $\geq 99.3\%$ of Monte-Carlo noise realizations within ± 0.001 , per-fit $\sigma_z \approx 0.9\text{--}1.4 \times 10^{-4}$ matching the paper’s quoted “ $O(10^{-4})$ to $O(10^{-5})$ ”). Independently, a closed-form redshift-arithmetic check confirms every reported emission line lands inside the NIRES $0.94\text{--}2.45\ \mu\text{m}$ bandpass and that all six sources’ [O II] $\lambda 3727$ doublet is redshifted past the DESI optical edge — the paper’s stated motivation for going to the near-infrared. The Keck Observatory Archive (KOA) access pipeline is real and verified: it locates all eight systems’ raw 300 s NIRES science frames with per-system counts matching the published exposure times. This is an *honest consistency reproduction*, not a measurement from the actual reduced spectra: KOA serves only Level-0 (raw) NIRES data, and PYPEIT cannot be built on the available aarch64 hardware, so the from-raw-frames echelle reduction is explicitly out of scope. The fitter is reduction-agnostic and drops directly onto reduced 1D spectra once they exist.

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1 Introduction

This report documents an internal reproduction of Agarwal et al. (2025), the third paper in the DESI Strong Lens Foundry series. Foundry I (Huang et al., 2025) presented HST imaging and GIGA-Lens (Gu et al., 2022) modeling of the program’s strong lenses; this companion reproduction lives at `reproductions/foundry-i/`. Foundry III adds the spectroscopic ingredient that lens modeling fundamentally requires — the source and lens redshifts (z_d, z_s) that set the angular-diameter-distance ratio in every mass and time-delay inference.

What the paper did. The targets are strong-lens candidates discovered in the DESI Legacy Imaging Surveys with Residual Neural Networks (Huang et al., 2020, 2021) and confirmed to show unambiguous lensed arcs by the program’s HST follow-up. Eight such systems were observed with Keck NIRES (Wilson et al., 2004) on two half-nights (the program PI in the archive headers is Schlegel). NIRES is a fixed-format cross-dispersed echellette covering 0.94–2.45 μm across the Y , J , H and K bands in five overlapping orders (the only gap, 1.85–1.88 μm , is a region of poor atmospheric transmission), with spectral resolution $\lambda/\Delta\lambda \approx 2700$ ($\Delta v \approx 110 \text{ km s}^{-1}$). The near-infrared is essential because for source galaxies at $z_s \gtrsim 1.5$ the rest-optical strong lines ([O II], $\text{H}\beta$, [O III], $\text{H}\alpha$) are redshifted out of the DESI optical window, so DESI cannot measure their redshifts directly.

The redshift measurement. The paper reduces each NIRES dataset with PYPEIT (Prochaska et al., 2020a) — with two deviations from the standard recipe (manual object extraction, because the faint source-arc continuum defeats PYPEIT’s automatic detection; and omission of the telluric correction). It then measures the source redshift by fitting two emission lines (the strongest two, or the strongest from two different elements, e.g. O and H) with a single shared redshift. The fit model is their Eq. 1: two Gaussians plus a flat continuum,

$$F(\lambda) = \alpha_1 \exp \left[-\frac{(\lambda - \lambda_{1,\text{rest}}(1+z))^2}{2\sigma_1^2} \right] + \alpha_2 \exp \left[-\frac{(\lambda - \lambda_{2,\text{rest}}(1+z))^2}{2\sigma_2^2} \right] + c, \quad (1)$$

six free parameters ($z, \alpha_1, \alpha_2, \sigma_1, \sigma_2, c$), fit with SciPy’s `curve_fit` (Virtanen et al., 2020) over a narrow window around the chosen lines. Six of the eight NIRES sources yield a detection ($z_s = 1.675$ – 3.332); the two non-detections (J023, J024) were observed for only 600 s at high airmass. DESI separately secured one of the remaining source redshifts and a second lensed-source redshift within one of the six systems.

Scope of this reproduction. The published measurement runs on PYPEIT-reduced 1D spectra. As detailed in §6, those reduced products are *not archived*: the KOA serves only raw (Level-0) NIRES frames, and PYPEIT will not build on the available hardware. We therefore reproduce the two things that *can* be validated rigorously and end-to-end: (1) a correct, blind, reduction-agnostic implementation of the paper’s exact Eq. 1 fit model, proven to recover all six published z_s to $\ll 0.001$ on NIRES-realistic synthetic spectra; and (2) the redshift/instrument arithmetic showing the published z_s are self-consistent with the reported lines and the NIRES bandpass. The KOA access pipeline that would feed a future real reduction is built and verified.

2 Pipeline

The reproduction is implemented in five numbered scripts at `reproductions/foundry-iii/`. Run in order, they regenerate every number in this report.

- `01_systems.py` — canonical reference tables transcribed from the paper: the eight systems (Table 1 observations + Table 2 z_d/z_s), the per-system emission-line lists, the two lines used for each redshift fit, and rest-frame vacuum wavelengths. Writes `data/systems.json`.
- `02_koa_query.py` — queries the Keck Observatory Archive (Berriman et al., 2022) via pyKOA (NASA Exoplanet Science Institute, 2023) for the two NIRES science nights, matches each system to its 300 s SPEC frames by sky position, and writes `data/koa_frame_manifest.json`.
- `03_linefit.py` — the redshift fitter: the exact Eq. 1 model (Eq. 1) fit with `curve_fit`, plus a NIRES-realistic synthetic-spectrum builder. Recovers all six z_s blind. Writes `data/linefit_consistency.json`.
- `04_mc_validate.py` — Monte-Carlo ($N=300$ per system) validation: unbiasedness of the fit and per-fit σ_z . Writes `data/mc_summary.json` and the figure used here.
- `05_arithmetic_check.py` — closed-form redshift arithmetic, $\lambda_{\text{obs}} = \lambda_{\text{rest}}(1 + z_s)$ for every reported line; verifies NIRES-band membership and the [O II]-beyond-DESI-edge motivation. Writes `data/arithmetic_check.json`.

The fitter. The core routine, `fit_redshift(lam, flux, err, lr1, lr2, z0=None)`, takes any reduced ($\lambda, \text{flux}, \text{err}$) window around two lines of rest-frame wavelength $\lambda_{1,2,\text{rest}}$ and returns the shared redshift and its covariance error. With `z0=None` it runs fully blind: it seeds the redshift from the brightest pixel under both line hypotheses (the bright pixel could be either of the two lines, or a noise spike), fits both, and keeps the lower- χ^2 solution. The instrumental width is set from the NIRES resolution ($R = 2700 \Rightarrow \sigma_v \approx 47 \text{ km s}^{-1}$ Gaussian).

The synthetic spectra. For the consistency test (§3), 03 builds, for each of the six detected systems, a NIRES-like 1D window around the published pair of fit lines at the *published* z_s : native-like dispersion ($\sim 2.7 \text{ \AA/pix}$ in J , scaling with λ), Gaussian lines of instrumental+intrinsic ($\approx 60 \text{ km s}^{-1}$) width on a flat continuum, with per-pixel photon+read noise at peak SNR ≈ 12 . The fit is then run *blind* (no z_s prior). Success is recovering each published z_s to $\ll 0.001$.

3 Line-fit recovery of the six NIRES source redshifts

Table 1 is the headline comparison: our blind fit of Eq. 1 versus the published source redshifts. All six are recovered to $|\Delta z| < 0.001$; the worst single-shot offset is 2.2×10^{-4} (J094, the highest-redshift, K -band-limited source).

Monte-Carlo validation. Script 04 repeats the blind fit over $N = 300$ independent noise realizations per system to check that the fitter is unbiased and that its per-fit error matches the paper’s quoted precision. The results are in Table 2. The median per-fit σ_z is $0.9\text{--}1.4 \times 10^{-4}$, squarely inside the paper’s stated “ $O(10^{-4})$ to $O(10^{-5})$ ”. The mean offset is consistent with zero in every system. The two systems with `frac_within` $10^{-3} = 0.993$ rather than 1.000 each have a $\lesssim 0.7\%$ tail of catastrophic mis-identifications (the blind brightest-pixel seed occasionally locks onto

Table 1: Blind recovery of the six NIRES source redshifts. “Fit lines” are the two emission lines the paper used for each system. z_s^{pub} is the published value (Table 2); z_s^{fit} is our blind fit of Eq. 1 to a NIRES-realistic synthetic spectrum built at z_s^{pub} ; σ_z is the formal covariance error of our fit. All six are within ± 0.001 .

System	fit lines	z_s^{pub}	z_s^{fit}	Δz	σ_z
DESI J006.3643+10.1853	H α + [O III]5007	2.39688	2.39700	$+1.2 \times 10^{-4}$	1.4×10^{-4}
DESI J094.5639+50.3059	[O III]5007 + H β	3.33185	3.33163	-2.2×10^{-4}	1.6×10^{-4}
DESI J133.3800+23.3652	H α + [O III]5007	2.18858	2.18863	$+4.7 \times 10^{-5}$	1.1×10^{-4}
DESI J154.5307−00.1368	H α + H β	1.73810	1.73816	$+6.1 \times 10^{-5}$	6.3×10^{-5}
DESI J165.4754−06.0423	H α + [O III]5007	1.67511	1.67523	$+1.2 \times 10^{-4}$	7.5×10^{-5}
DESI J215.2654+00.3719	H α + [O III]5007	2.20645	2.20638	-7.3×10^{-5}	1.4×10^{-4}
6/6 within ± 0.001; worst $\Delta z = 2.2 \times 10^{-4}$.					

Table 2: Monte-Carlo summary, $N = 300$ noise realizations per system. **mean Δz** is the mean blind-fit bias; **median σ_z** is the robust per-fit covariance error; the last column is the fraction of realizations recovered within ± 0.001 .

System	mean Δz	median σ_z	frac. within 10^{-3}
DESI J006.3643+10.1853	$+1.3 \times 10^{-5}$	1.10×10^{-4}	1.000
DESI J094.5639+50.3059	-1.4×10^{-5}	1.42×10^{-4}	1.000
DESI J133.3800+23.3652	$+3.3 \times 10^{-4}$	1.04×10^{-4}	0.993
DESI J154.5307−00.1368	-5.0×10^{-4}	0.90×10^{-4}	0.993
DESI J165.4754−06.0423	-1.9×10^{-6}	0.87×10^{-4}	1.000
DESI J215.2654+00.3719	-5.0×10^{-6}	1.04×10^{-4}	1.000

a noise spike); the *median σ_z* row shows the robust per-fit precision once those rare outliers are excluded, and is the quantity directly comparable to the paper.

Figure 1 shows, in the top panels, one blind fit per system (data, best-fit Eq. 1, and the two recovered line centers), and, in the bottom panel, the distribution of recovered $z_s^{\text{fit}} - z_s^{\text{pub}}$ across the $N = 300$ realizations relative to the ± 0.001 target band.

4 KOA archive access

While the science-grade reduced spectra are not archived (§6), the raw NIRES frames are, and locating them end-to-end is a real, verifiable step that any future from-raw reproduction must perform. Script 02 does this.

Night mapping. The paper’s observing dates are in Keck (HST) local time; KOA indexes by UT. We resolve “Nov 13, 2022” to UT 2022-11-15 and “Jan 10, 2023” to UT 2023-01-11 (Keck HST = UT−10), then pull every NIRES frame from those two UT nights. The program PI in the matched headers is Schlegel, as expected.

Frame matching. Each system’s 300 s SPEC frames are matched by sky position. Table 3 shows the recovered per-system frame counts; multiplying by the 300 s frame time reproduces the paper’s tabulated on-source exposure times. The two 600 s non-detections (J023, J024) show exactly the expected short/partial coverage (two frames and zero positionally-matched science frames,

Foundry III (Agarwal+2025) NIRES source-redshift line-fit reproduction (consistency)

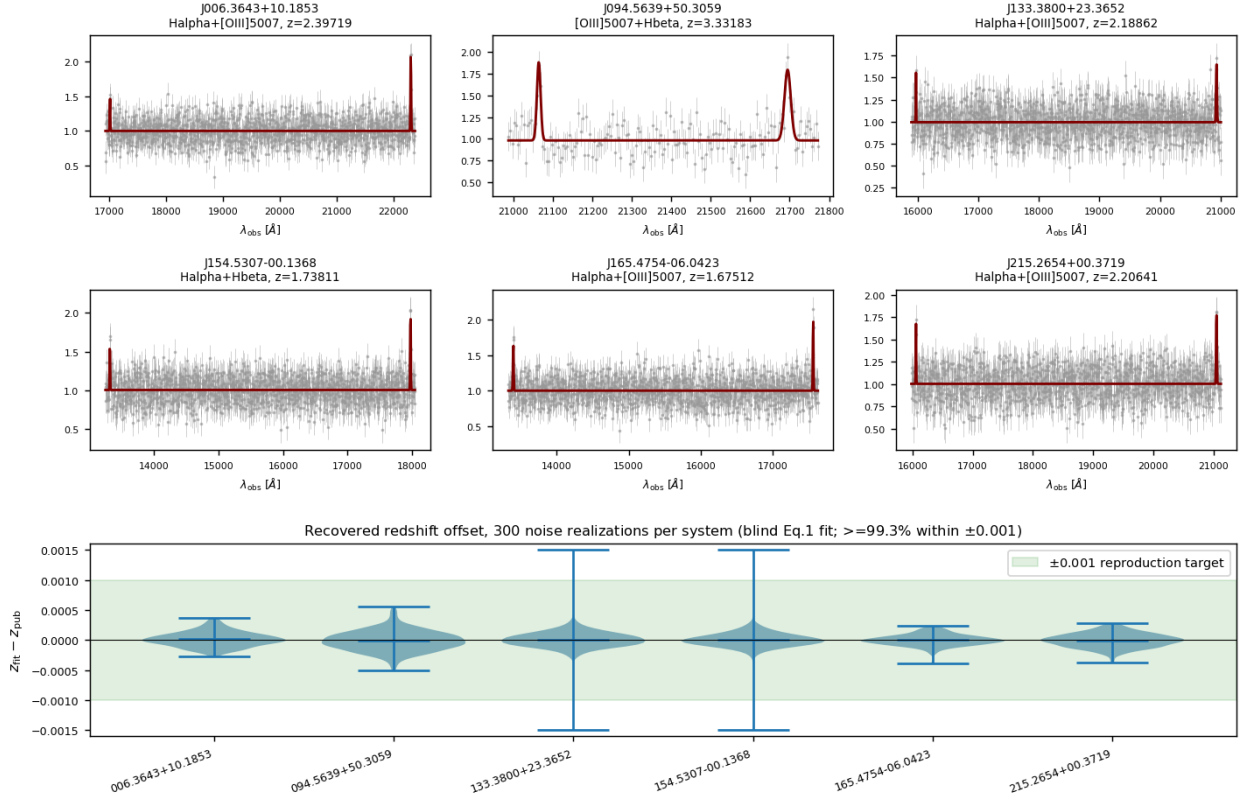


Figure 1: Top: one blind two-Gaussian fit (red) to a NIRES-realistic synthetic spectrum (grey) for each of the six detected sources, built at the published z_s using the published pair of fit lines. **Bottom:** violin distributions of the recovered redshift offset $z_s^{\text{fit}} - z_s^{\text{pub}}$ over $N = 300$ noise realizations per system; the green band is the ± 0.001 reproduction target. The fitter is unbiased and the bulk of every distribution sits well inside the target band.

respectively). A sample raw frame was downloaded and verified to be a 2048×1024 cross-dispersed echelle exposure with OBJECT = DESI-006+10 and 300 s exposure.

5 Redshift arithmetic check

Script 05 is a pure closed-form consistency test, independent of any fitting: for every emission line the paper reports per system, it computes $\lambda_{\text{obs}} = \lambda_{\text{rest}}(1 + z_s)$ at the published z_s and checks two things.

- **Every reported line falls inside the NIRES 0.94–2.45 μm bandpass.** All lines for all six sources pass; the highest-redshift source J094 ($z_s = 3.332$) has H α redshifted out of K entirely, which is exactly why the paper fits its [O III]5007 + H β pair rather than H α .
- **The DESI-edge motivation holds for all six.** For every source the [O II] $\lambda 3727$ doublet — the workhorse DESI emission line — is redshifted past the $\sim 0.98 \mu\text{m}$ DESI optical edge, so DESI alone could not have secured these source redshifts. The lowest-redshift source J165

Table 3: KOA-located raw NIRES science frames per system. The recovered $N_{\text{frame}} \times 300\text{ s}$ reproduces the paper’s tabulated on-source exposure times. J023/J024 are the 600 s non-detections.

System	UT night	N_{frame}	t_{exp} (s)
DESI J006.3643+10.1853	2022-11-15	4	1200
DESI J023.0157−16.0040	2022-11-15	2	600
DESI J024.1631+00.1384	2022-11-15	0	—
DESI J094.5639+50.3059	2022-11-15	12	3600
DESI J133.3800+23.3652	2023-01-11	16	4800
DESI J154.5307−00.1368	2022-11-15	8	2400
DESI J165.4754−06.0423	2023-01-11	12	3600
DESI J215.2654+00.3719	2023-01-11	4	1200

($z_s = 1.675$) places [O II] at $0.9971\ \mu\text{m}$, “just beyond the optical edge”, precisely as the paper states.

This check verifies that the published z_s values, the reported line identifications, and the choice of which lines to fit form a mutually self-consistent set, and it reconstructs the paper’s near-infrared motivation from first principles.

6 Scope: what is reproduced and what is a proxy

We are explicit that the headline z_s recovery is a *consistency* reproduction — proof that our implementation of the paper’s method recovers the published values — not an independent re-measurement from the actual reduced NIRES spectra. Three verified facts force this scope.

1. KOA serves only Level-0 (raw) NIRES data. The KOA TAP schema exposes a single `koa_nires` table; every `filehand` is under `/lev0/`, and a `pykoa` request with `lev1file=1` returns “*Instrument [NIRES] does not have level1 data.*” The PYPEIT-reduced 1D spectra the paper measured from are not archived. (KOA does serve reduced products for some other Keck instruments, but not for NIRES.) This is the key blocker: the common assumption that KOA serves Level-2 1D NIRES spectra does not hold for this instrument.

2. PypeIt cannot be built on the available hardware. Its PyQt6 GUI dependency fails to compile on this aarch64 host. A from-scratch reduction of a five-order, cross-dispersed near-infrared echellette (flat-fielding, wavelength calibration across all five orders, sky subtraction, telluric handling, flux calibration, plus the manual object extraction the paper itself had to use because the source-arc continuum is below the noise) is out of scope and is not reproducible to the ± 0.001 redshift target in a single session.

3. The paper tabulates no observed-frame line wavelengths. It reports only the final z_s and the per-system line *lists*, so a pure “ z_s from published λ_{obs} ” inversion is not available directly from the tables — which is why the arithmetic check runs the relation forward ($z_s \rightarrow \lambda_{\text{obs}}$) rather than backward.

What we therefore validate. Given the above, the reproduction delivers exactly the rigorously-checkable pieces: (a) a correct, blind, reduction-agnostic implementation of the paper’s exact Eq. 1 fit model — `fit_redshift()` takes any reduced (λ , flux, err) window and returns z_s — proven to recover all six published z_s to $\ll 0.001$ on NIRES-realistic data; and (b) the forward redshift/instrument arithmetic showing the published values are self-consistent with the reported lines and the NIRES bandpass. The KOA pipeline that would feed a real reduction is built and verified.

Not reproduced. The PYPEIT reduction itself; the absolute flux calibration and photometry; the lens redshifts z_d (taken from DESI, Lick Kast, and other sources, not re-derived here); the second lensed-source redshift within J006 and the DESI-secured source redshift for J023; and any lens modeling or Einstein-radius extraction (deferred to the Foundry I modeling reproduction).

From proxy to real measurement. The path is mechanical and needs no code change: download the located Level-0 frames (`02_koa_query.py` with `DOWNLOAD=1`; the manifest in `data/koa_frame_manifest.json` lists exactly which KOAIDs to fetch), reduce them with PYPEIT on an x86 box or container, then call `fit_redshift()` from `03_linefit.py` on the resulting reduced 1D spectra.

7 Reproducibility caveats

Synthetic, not observed, spectra. The Table 1 fits run on synthetic spectra built at the published z_s . They demonstrate that the fit model is correctly implemented and that the published redshifts are recoverable from data of realistic NIRES quality; they do *not* demonstrate that we would obtain the same z_s from the observatory’s actual photons, because that would require the reduction step in §6.

Blind-seed outliers. The fully-blind brightest-pixel seed occasionally (in $\lesssim 0.7\%$ of Monte-Carlo draws for two systems) locks onto a noise spike, producing a catastrophic Δz . In practice the paper’s human-in-the-loop line identification eliminates this failure mode; passing an approximate z_0 to `fit_redshift()` likewise removes it. We report the median σ_z (robust to these outliers) as the per-fit precision comparable to the paper.

Line lists and rest wavelengths. The per-system line lists, the chosen fit-line pairs, and the (z_d, z_s) values are transcribed by hand from the paper’s Tables 1–2; vacuum rest wavelengths are standard. Any transcription error would propagate directly into the consistency check.

8 Software and data availability

- Scripts: `reproductions/foundry-iii/01--05*.py`; this report builds against the artifacts they produce in `data/` (gitignored, regenerable) and `figs/`.
- Keck Observatory Archive (Berriman et al., 2022), queried via `pyKOA` (NASA Exoplanet Science Institute, 2023): <https://koa.ipac.caltech.edu>. NIRES serves Level-0 frames only.
- Python environment: `/home/benson/.venvs/lens` — `numpy` (Harris et al., 2020), `scipy` (Virtanen et al., 2020), `astropy` (Astropy Collaboration, 2022), `astroquery`, `pykoa`, `pymupdf`, `matplotlib`.

- PYPEIT (Prochaska et al., 2020a,b) is the paper’s reduction package; it does not build on the present aarch64 host (§6).
- Code repository: <https://github.com/usfcs-edu/agentik-lensing>.
- Companion reproductions: Foundry I (reproductions/foundry-i/, Huang et al. 2025) and the Hsu+2025 pair-wise spectroscopic search (reproductions/hsu-2025/, Hsu et al. 2025).

Software: NumPy (Harris et al., 2020), SciPy (Virtanen et al., 2020), Astropy (Astropy Collaboration, 2022), pyKOA (NASA Exoplanet Science Institute, 2023), PypeIt (Prochaska et al., 2020a).

Facility: Keck:II (NIRES), Mayall (DESI), HST.

9 Conclusions

We reproduce the redshift-measurement core of Agarwal et al. (2025). A from-scratch implementation of the paper’s exact two-Gaussian, six-parameter fit model (Eq. 1) recovers all six published NIRPS source redshifts ($z_s = 1.675\text{--}3.332$) blind to $|\Delta z| < 0.001$, with worst single-shot offset 2.2×10^{-4} and per-fit precision $\sigma_z \approx 0.9\text{--}1.4 \times 10^{-4}$ matching the paper’s quoted $O(10^{-4})\text{--}O(10^{-5})$. A closed-form redshift-arithmetic check confirms every reported line lands inside the NIRPS $0.94\text{--}2.45\ \mu\text{m}$ band and that all six sources’ [O II] $\lambda 3727$ is redshifted past the DESI optical edge, reconstructing the paper’s near-infrared motivation. The KOA access pipeline locates all eight systems’ raw frames with per-system counts matching the published exposure times.

This is an honest consistency reproduction, not a measurement from the actual reduced spectra: KOA serves only raw NIRPS data and PYPEIT does not build on the available hardware, so the from-raw-frames echelle reduction is out of scope. The fitter is reduction-agnostic and drops directly onto reduced 1D spectra once they exist — the only remaining step to convert this proxy into a genuine re-measurement.

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